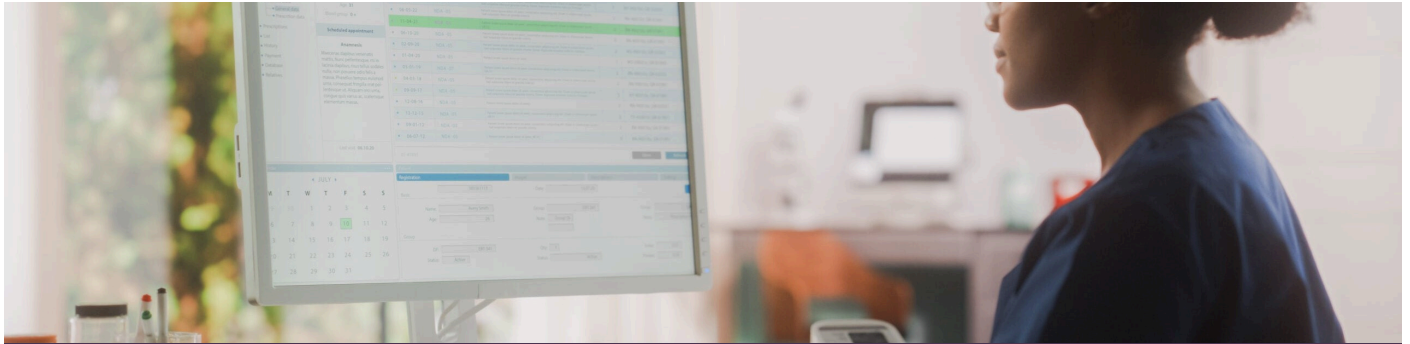


## 07 | OVERVIEW



## Rapid Response to Heat Waves: Using Electronic Health Records to Protect Patients in Near-Real Time

### Abstract

Public health officials often wait months to learn who was harmed during a heat wave. A 2026 study led by Harvard Chan C-CHANGE and Boston University showed that electronic health records can reveal heat's health impacts within about a week. By analyzing 8.1 million emergency department visits, researchers found that roughly 1 in 10 visits on extreme heat days were caused by high temperatures.

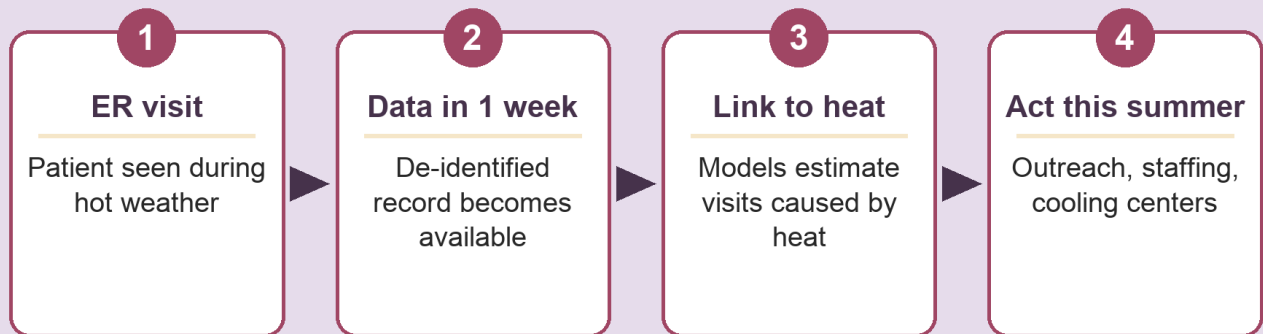
### Introduction

As summer heat waves grow longer and more dangerous, knowing who is getting sick, and when, can save lives. Traditionally, studies of heat and health have relied on insurance claims data, which often arrive 6 to 18 months after an extreme heat event. By then, the summer is over, and it is too late to adjust warnings, staff hospitals, or open cooling centers for the people most affected. Researchers at the Harvard T.H. Chan School of Public Health and Boston University School of Public Health, working with the health data company Truveta, tested a faster approach: using de-identified electronic health records to measure heat's toll while the heat season is still underway.

## How Do Scientists Study This?

The team analyzed 8.1 million emergency department (ED) visits from the summers of 2023 to 2025, using de-identified electronic health records from health systems across 19 Eastern U.S. states. They linked each day's maximum temperature to state-level counts of ED visits for all causes, heat-related illnesses, kidney diseases, and psychiatric conditions. Using statistical models, they estimated how much ED visits increased on hotter days. They then calculated how many visits during the June and August 2025 heat waves could be attributed to extreme temperatures. Crucially, these records were available within one week of each visit, rather than many months later.

### From health records to rapid action



#### Collect records

De-identified emergency visit records from 19 Eastern states became available within about one week.

#### Link temperature

Daily maximum temperatures were matched to counts of visits for heat illness, kidney disease, and mental health.

#### Estimate heat's toll

Statistical models calculated how many extra visits during heat waves were caused by high temperatures.

## What Did They Discover?

During the June 2025 heat wave, about 9.2% of all emergency visits, roughly 6,986 extra visits, were attributable to heat. During the August heat wave, about 8.3%, or 6,351 extra visits, were attributable to heat. Together, the two heat waves caused over 13,000 excess ED visits. Nearly 30% of kidney disease visits during heat waves were linked to heat. Across the whole warm season, hotter temperatures were associated with higher visit rates for all causes (6.3%), kidney diseases (19.5%), and psychiatric conditions (8.8%). Surprisingly, heat-related visits declined only modestly between June and August, suggesting people did not adapt much as summer went on.

### Heat and emergency visits

**8.1M**

emergency visits analyzed, 2023 to 2025

**13,000+**

extra visits caused by two 2025 heat waves

**~30%**

of kidney disease visits during heat waves linked to heat

Lead author Amruta Nori-Sarma called the study a proof of concept for turning near-real-time health records into actionable intelligence during an ongoing climate event.

## Takeaways

### Heat sends people to the ER

About 1 in 10 emergency visits on extreme heat days were caused by high temperatures.

### Kidneys at risk

Nearly 30% of kidney disease emergency visits during heat waves were linked to heat.

### Speed saves lives

Week-old data lets health systems adjust warnings, staffing, and programs in the same summer.

# Key Terms

## Electronic health record (EHR)

A digital version of a patient's medical history kept by health care providers.

## De-identified data

Health information with names and other personal details removed to protect patient privacy.

## Attributable visits

The number of health care visits estimated to be caused by a specific factor, such as extreme heat.

## Surveillance

In public health, the ongoing collection and analysis of health data to detect and respond to problems.

## Heat action plan

A community plan for protecting people during extreme heat, such as issuing warnings and opening cooling centers.

### Match-Up: draw a line from each term to its clue

**A** Electronic health record (EHR) •

**B** De-identified data •

**C** Attributable visits •

**D** Surveillance •

**E** Heat action plan •

• **1** A community plan for protecting people during extreme heat

• **2** A digital version of a patient's medical history kept by health care...

• **3** In public health, the ongoing collection and analysis of health data to...

• **4** Health information with names and other personal details removed to protect patient...

• **5** The number of health care visits estimated to be caused by a...

Answer key: A-2 · B-4 · C-5 · D-3 · E-1

## 07 | ACTIVITY

## Activity

### Heat Wave Command Center

Students act as a public health team receiving weekly emergency visit data and decide how to respond during a heat wave.

Materials: Data cards with the study's key numbers; Map of a sample city; Response option cards: cooling centers, outreach, staffing, messaging

1. Review the key findings and identify the groups most at risk during heat waves.
2. Imagine you receive data one week into a heat wave showing a spike in kidney disease visits. Choose two responses and justify them.
3. Compare this with receiving the same data 12 months later. What could you no longer do?
4. Design one message for older adults or people with kidney disease, warning them about heat.
5. Explain how you would use the next week's data to check whether your response is working.

## References

Miller, A. (2026). Rapid response to heat waves: Using electronic health records to protect patients in near-real time. Harvard T.H. Chan School of Public Health.  
<https://hsph.harvard.edu/climate-health-c-change/news/rapid-response-to-heat-waves-using-electronic-health-records-to-protect-patients-in-near-real-time/>

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